THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 2: **Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a chromatogram of an ink with R_f values 0.3, 0.6, 0.9, the component most soluble in the solvent is: 1

(a) the one with R_f 0.9 **(b)** the one with R_f 0.3
(c) the one with R_f 0.6 **(d)** all equal

2. Which of the following separates a mixture using the difference in boiling points? 1

(a) filtration **(b)** winnowing **(c)** sieving **(d)** distillation

3. Assertion (A): Crystallisation is preferred to evaporation for obtaining pure solids from a solution. Reason (R): Crystallisation removes soluble impurities that remain in the liquid. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. A 100 mL solution of 20 per cent (m/v) salt solution is diluted to 400 mL. The new strength is: 1

(a) 80 per cent **(b)** 10 per cent **(c)** 5 per cent (m/v) **(d)** 20 per cent

5. A sample of water boils at 102 degrees Celsius at normal pressure. This indicates: 1

(a) a gas **(b)** pure water
(c) dissolved impurities **(d)** an alloy

6. Assertion (A): A pencil line, not ink, is used in chromatography. Reason (R): Graphite does not dissolve in the solvent. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. 30 mL of acetone is mixed with water to prepare a 6 per cent (v/v) solution. The final volume is: 1

(a) 180 mL **(b)** 500 mL **(c)** 50 mL **(d)** 300 mL

8. A student separates two components A and B by shaking with a solvent in which only A is soluble. This method is called: 1

(a) chromatography only **(b)** winnowing
(c) sublimation **(d)** solvent extraction

9. **Assertion (A):** A suspension is a heterogeneous mixture. 1
Reason (R): The particles of a suspension do not settle down when the mixture is left undisturbed. Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A true solution scatters a beam of light passing through it. 1
Reason (R): The solute particles of a true solution are very small (diameter less than 1 nm).
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the volume of a solution not equal the sum of the volumes of solute and solvent in some mixtures like alcohol and water? 2

12. Why does a separating funnel work only for immiscible liquids? 2

13. Calculate the volume of ethanol in 750 mL of a mixture that is 12 per cent (v/v). 2

14. A mixture shows a path of light, settles on standing and is filtered. Classify it and justify. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how a mixture of sand, sugar and chalk powder can be separated. 3

16. Explain why the same solvent can produce different separation results for different mixtures. 3

17. Why is a thin-layer or paper chromatography of a pure compound expected to show a single spot while that of a mixture shows several? Explain with an example of black ink. 3

OR

Explain why a mixture of two liquids with boiling points 78 and 82 degrees Celsius cannot be separated well by simple distillation. 3

18. State three differences between sublimation and evaporation with an example each. 3

OR

Name the separation methods in each step of making sugar from sugarcane: crushing, filtering, evaporation, crystallisation, centrifuge. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In the salt pans of Gujarat, sea water is let into shallow fields and left in the sunshine. After many days white crystals of salt are collected. The crude salt is later dissolved in water, filtered and crystallised again to make it purer. 5
(i) Name the process by which water is removed in the salt pans.
(ii) Why can salt not be separated from sea water by filtration?
(iii) Name the method used to purify the crude salt crystals.
(iv) Why is sunshine used for evaporation instead of boiling the water to dryness?

20. A crude oil sample is separated into fractions at 40-70, 70-120, 120-170, 170-250, above 250 degrees Celsius. Name the likely fractions and uses and explain the use of the fractionating column. 5

OR

Design an experiment to test the claim that 'a pure compound has a sharp melting point while a mixture melts over a range' with naphthalene and a naphthalene-sulphur mixture. 5

END OF QUESTION PAPER — Total: 40 Marks